

Education

- Ph.D. Mechanical Engineering — ETH Zurich 2016 – 2022
Simulation and control of artificial microswimmers in blood - Advisor: Petros Koumoutsakos
- MSc Computational Science and Engineering — EPFL 2013 – 2016
- BSc Physics — EPFL 2010 – 2013

Academic Appointments

- Lecturer, Harvard University 2025 – present
- Research Associate, Harvard University 2022 – present
- Postdoctoral Fellow, Harvard University 2022 – 2023

Teaching Experience

Lecturer

- **Advanced Scientific Computing: Stochastic Methods for Data Analysis, Inference and Optimization** - Harvard University — Graduate course (58 students) Fall 2025
- **High Performance Computing for Science and Engineering I & II** - ETH Zurich — Co-lecturer (160 students) 2018–2020

Teaching Assistant

- Stochastic Methods for Data Analysis, Inference and Optimization - Harvard University 2021–2023
- High Performance Computing for Science and Engineering I & II - ETH Zurich 2017
- Models, Algorithms and Data: Introduction to Computing - ETH Zurich 2017
- General Physics - University of Lausanne 2015
- Introduction to Programming - EPFL 2014

Supervision

- Supervised 10 students for semester, Bachelor, and Master theses

Academic Service

- Reviewer for *Biophysical Journal*, *Advanced Intelligent Systems*, *Physics of Fluids*, and others
- Co-organizer, *Widely Applied Math Seminar*, Harvard SEAS 2022

Grant Applications

- SALATA: Sustainable Building and Urban Future 2025
- AWS/HDSI: Physics-aware foundation models for extreme atmospheric events — \$197k 2025
- EuroHPC: Targeted Drug Delivery with Microswimmers — 150k node hours 2021
- CHRONOS: Simulation of Microfluidics for Mechanical Cell Separation — 2M node hours 2016

Software

- **Mirheo** — C++/CUDA/MPI framework for microfluidics simulations
- **Korali** — C++/OpenMP/MPI framework for large-scale Bayesian inference

Publications

Full list on my [google scholar profile](#).

- Amoudruz, L., Litvinov, S., Papadimitriou, C., Koumoutsakos, P. (2026) *Bayesian inference for PDE-based inverse problems using the optimization of a discrete loss*, CMAME, 455, 118903.
- Amoudruz, L., Litvinov, S., Koumoutsakos, P. (2025). *Path planning of magnetic microswimmers in high-fidelity simulations of capillaries with deep reinforcement learning*. Physics of Fluids, 37, 071703.
- Amoudruz, L., Karnakov, P., Koumoutsakos, P. (2025). *Contactless transport of particles using hydrodynamics*. Journal of Fluid Mechanics, 1014:A15.
- Alexeev, D., Litvinov, S., Economides, A., Amoudruz, L., Toner, M., Koumoutsakos, P. (2025). *Inertial focusing of spherical particles: The effects of rotational motion*. Physical Review Fluids, 10(5), 054202.
- Karnakov, P., Amoudruz, L., Koumoutsakos, P. (2025). *Closed-loop control and navigation in microfluidics through optimizing a discrete loss*. Physical Review Letters, 134(4), 044001.
- Amoudruz, L., Economides, A., Koumoutsakos, P. (2024). *The volume of healthy red blood cells is optimal for advective oxygen transport in arterioles*. Biophysical Journal, 123(10), 1289–1296.
- Amoudruz, L., Economides, A., Arampatzis, G., Koumoutsakos, P. (2023). *The stress-free state of human erythrocytes: Data-driven inference of a transferable RBC model*. Biophysical Journal, 122(8), 1517–1525.
- Amoudruz, L., Koumoutsakos, P. (2022). *Independent control and path planning of microswimmers with a uniform magnetic field*. Advanced Intelligent Systems, 4(3), 2100183.
- Economides, A., Arampatzis, G., Alexeev, D., Litvinov, S., Amoudruz, L., et al. (2021). *Hierarchical Bayesian uncertainty quantification for a model of the red blood cell*. Physical Review Applied, 15(3), 034062.
- Alexeev, D., Amoudruz, L., Litvinov, S., Koumoutsakos, P. (2020). *Mirheo: High-performance mesoscale simulations for microfluidics*. Computer Physics Communications, 254, 107298.
- Wälchli, D., Martin, S.M., Economides, A., Amoudruz, L., et al. (2020). *Load balancing in large-scale Bayesian inference*. PASC Conference.
- Economides, A., Amoudruz, L., Litvinov, S., Alexeev, D., et al. (2017). *Towards the virtual rheometer: High performance computing for the red blood cell microstructure*. PASC Conference.

Invited Talks

- *Optimal Path Planning at the Microscale for Artificial Microswimmers and Cellular Manipulation*, Harvard Rising Stars in Computational Science and Engineering, Cambridge, MA, USA
- *Data-driven inference of a transferable red blood cell model*, CMBE 2024, Washington, DC, USA
- *Path planning of swimmers in complex flows with reinforcement learning*, APS-DFD 2023, Washington, DC, USA
- *Hierarchical Bayesian inference for a red blood cell model*, SIAM-UQ 2022, Atlanta, USA
- *Magnetic navigation of artificial bacteria flagella in blood and water*, APS-DFD 2021, Phoenix, USA
- *Fingering instability in a Hele-Shaw cell*, PASC 2018, Basel, Switzerland